The Joux–Vitse sieve on E(F_{p^6}) against a strong rho

Weak class only (never prime order); the cover's Jacobian has genus 3. Study `research/jv_sieve_strong_rho_20261019`, run 2026-10-19/20, protocol frozen before data. Every number is copied from `summary.json`, computed by `examples/jv_sieve_study.rs`.

Scope: a weak class only. Curves `y² = h(x)(x−α)(x−σα)` over `F_{p⁶}` with full rational 2-torsion whose isogeny class contains a weak curve (about 55–63 % of such curves, ledger §18.5). The class never has prime order. The cover's Jacobian has **genus 3**. Nothing here is a statement about a prime-order or generic curve.

Class: measurement of an existing algorithm (the sieve of `src/cryptanalysis/jv_sieve.rs`) against a stronger reference than the ledger had. Not a new algorithm. Protocol frozen before any data: `PROTOCOL.md` (deviations log appended at its end). Numbers: `TABLES.md`, `summary.json` raw runs: `raw/` report: `REPORT.pdf`.

What was measured

Instances `generate_spec(p, seed)`, seeds 1–16, `p ∈ {251, 503, 1009, 1511, 1823}` (`ℓ ≈ 2^45.8 … 2^63.0`; 1823 is the last `p` with `ℓ < 2^63`). The sieve and every rho arm ran on the same instances and host. Every run was checked against the planted logarithm: all sieve runs (80 per `JV_LIN` arm) and all 268 rho runs recovered it.

- **Sieve:** `jv_cover --exp sieve --stop 64`, dense and sparse FGLM descent, paired.
- **Rho:** `src/cryptanalysis/jv_rho_strong.rs`, a 2×2 factorial: R0 plain, R1 batched inversion (k = 16), R2 negation map, R3 negation
1. batched (**the strong rho**), all with 256 multipliers and

  distinguished points. Costs are *counted by operation class* and priced twice: in the **harness unit** (`F_{q³}` product 55, square 55, inversion 166 `F_p` multiplications, measured from the harness's own `Fq3`; one affine addition is 331) and in a **tuned unit** (Karatsuba, Chung–Hasan squaring, adjugate inversion; 24/18/62) that credits rho alone.
- `S = F_p multiplications / (331·√ℓ)`. S^+ also charges every `F_p` addition, lookup and native operation one multiplication, on both sides.

Results

1. The sieve still beats a strong rho, from about p = 1009 up,

and the gap widens. Paired sieve/rho ratios, 95 % bootstrap (n = 16; below 1 is the sieve cheaper):

p	S, harness	S, tuned	S^+, harness	S^+, tuned
251	18.5 [5.4, 47]	44 [13, 109]	37 [12, 89]	73 [23, 176]
503	0.48 [0.35, 0.64]	1.15 [0.84, 1.53]	1.34 [0.94, 1.82]	2.65 [1.85, 3.62]
1009	0.024 [0.019, 0.031]	0.058 [0.044, 0.074]	0.067 [0.053, 0.084]	0.132 [0.105, 0.165]
1511	0.0090 [0.0070, 0.012]	0.022 [0.017, 0.029]	0.028 [0.022, 0.036]	0.055 [0.044, 0.071]
1823	0.0042 [0.0032, 0.0058]	0.010 [0.0078, 0.014]	0.015 [0.0125, 0.020]	0.030 [0.025, 0.038]

The pre-registered kill rules did not fire: K1 (S^+ ratio interval reaching 1 at any `p ≥ 1009`, harness price) and K2 (the same in the tuned price or inside the §5a band) both hold at 1009, 1511 and 1823. Against the cheapest rho in the tuned price (R2, post hoc) the 1009 ratios are 0.062 [0.045, 0.082] (S) and 0.137 [0.104, 0.183] (S^+). The distance remaining is therefore a factor of about 65 (S^+, harness) to 33 (S^+, tuned) at `p = 1823`.

2. The crossover moves up against a strong rho.

The ledger's crossover against the weak rho was `p ≈ 333–430`. Against the strong rho it is between `p = 251` and `p ≈ 503` in S (harness) and unresolved at 503 in S^+ or the tuned price, resolved by 1009.

3. §5a band.

Native work priced at 0, 0.1 and 1 GAE per operation on every arm, rho included, two layers (A: all non-multiplication work including `F_p` additions; B: without them), the sieve side carrying an upper bound for the F4/linear-algebra additions it does not emit (T2). The sieve ranks below rho at every weight and in all four columns at `p = 1009, 1511, 1823` (layer A, `w = 1`, tuned: 0.37 [0.29, 0.46], 0.16 [0.13, 0.21], 0.088 [0.072, 0.11]). At `p = 251` rho ranks below; at `p = 503` the order changes inside the band, so the pair is **unresolved** there.

4. The ℓ^{1/3} claim is not supported over the measured range

(K3 fires). Fitted exponent of total `F_p` multiplications in ℓ, 251–1823: sieve −0.122 [−0.165, −0.084] against the theory's 1/3; strong rho 0.508 [0.480, 0.538] against 1/2 (the instrument check passes). Post hoc, over `p ≥ 1009` (constant `m = 9`): sieve 0.052 [−0.015, 0.118], excluding 1/3. The sieve's cost falls from `p = 251` to `1009` (the sieve runs at `m = 10` at 251–503 and `m = 9` from 1009, and relations per line grow with `p`) and is then about flat, between `2^{31.6}` and `2^{31.9}` multiplications. It beats rho because rho grows as `ℓ^{1/2}` while the sieve does not grow over this range, **not** because it follows `ℓ^{1/3}`. Whether the cost starts to rise like `ℓ^{1/3}` beyond `p = 1823` is untested here (ℓ exceeds 64 bits next), so extrapolating on `ℓ^{1/3}` is an extrapolation with no measured exponent behind it.

5. What the rho accelerations buy in this tower

(R0 ⟶ R2 ⟶ R3, T3): *Negation* is worth about 1.2–1.6× (theory √2). *Batched inversion buys nothing*: an inversion costs about 3 products here, so Montgomery's trick (3 products) is neutral in the harness price and slightly worse in the tuned price (R1 0.542 against R0 0.499 at `p = 1009`). The brief's "2–3× cheaper per step" comes from cheaper field arithmetic (tuned price: 2.4×), not from batching. Positive control: the plain arm gives S = 1.14–1.45, the ledger's reference ≈ 1.3. A/A at `p = 1009` (new walk randomness): R0 1.25 against 1.29, R3 0.823 against 0.808, inside their intervals; the sieve's counts repeat identically on all 16 instances.

6. Sparse FGLM in the sieve's descent

(T4, PR #2244's linear stage, `JV_LIN`): the descent rows are bit-identical in 80/80 dense/sparse pairs (same digest and outcome). The linear stage is 2.08–2.20× cheaper, descent multiplications 1.28–1.29×, and the end-to-end `S` 1.014× (251) to 1.126× (1009, 1511) cheaper, 1.096× at 1823 (intervals in T4). The sparse path fell back to the dense one 0.4–7.5 times per run; that cost is counted. This is an engineering gain on a route that already beats rho; it moves no ranking.

Limits

- Weak class only; `p ≤ 1823` (`ℓ < 2^63`). Harness unit throughout; wall time is an observation (sieve 4 threads, rho one core: at `p = 1823` the sieve's mean wall is 9 s against 760 s for one strong-rho run; not a speedup claim).
- The isogeny walk to reach the class (ledger §13, §18.4) was **not re-measured** end-to-end from a non-weak curve against the strong rho is not asserted here.
- The tuned price is a cost model, not a faster implementation; rho's native work (canonicalisation, hashing, DP checks, scalar updates, cycle-window comparisons) is counted and priced in S^+ and the band. The sieve's F4 additions are not emitted by the harness (implementation gap): the band uses an upper bound, labelled as such.
- The post hoc fit and the R2 comparison are labelled post hoc and were not in the protocol.

Figures

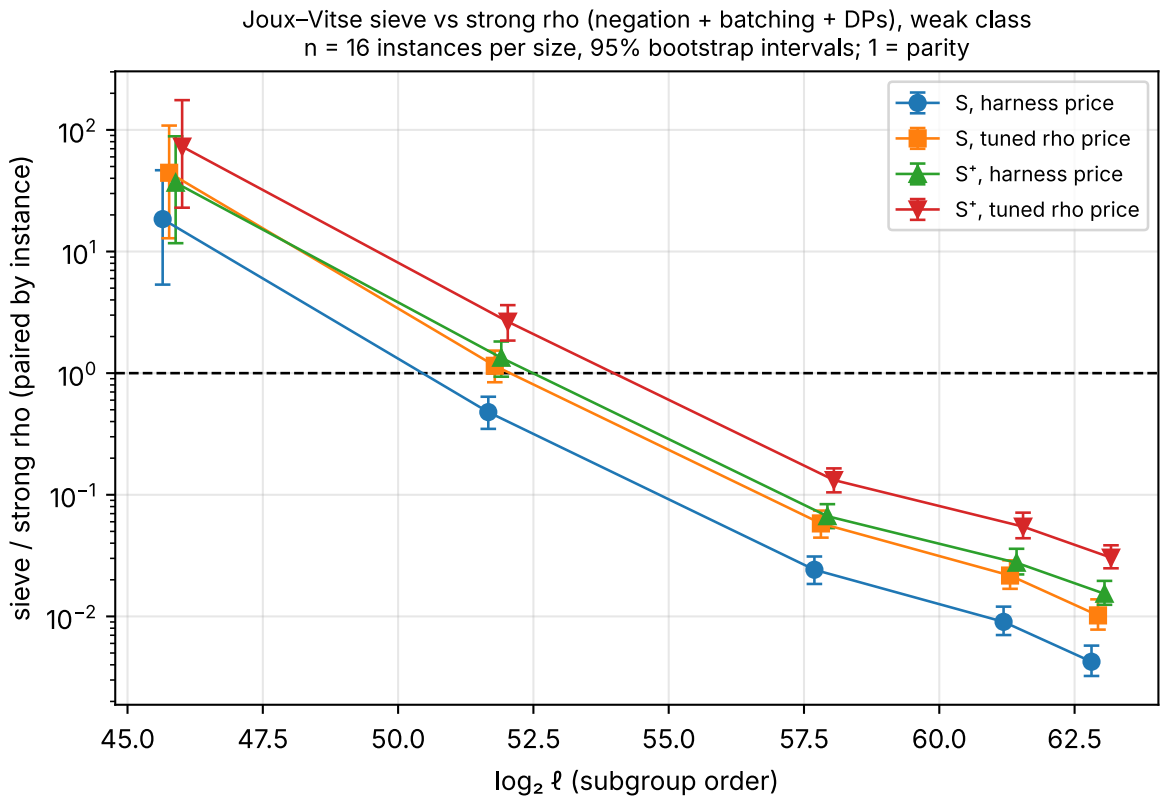


Figure 1. Sieve over strong rho (R3), paired by instance, four prices; 1 is parity. Study jv_sieve_strong_rho_20261019, instances seeds 1–16 per size, 95% bootstrap intervals; source summary.json.

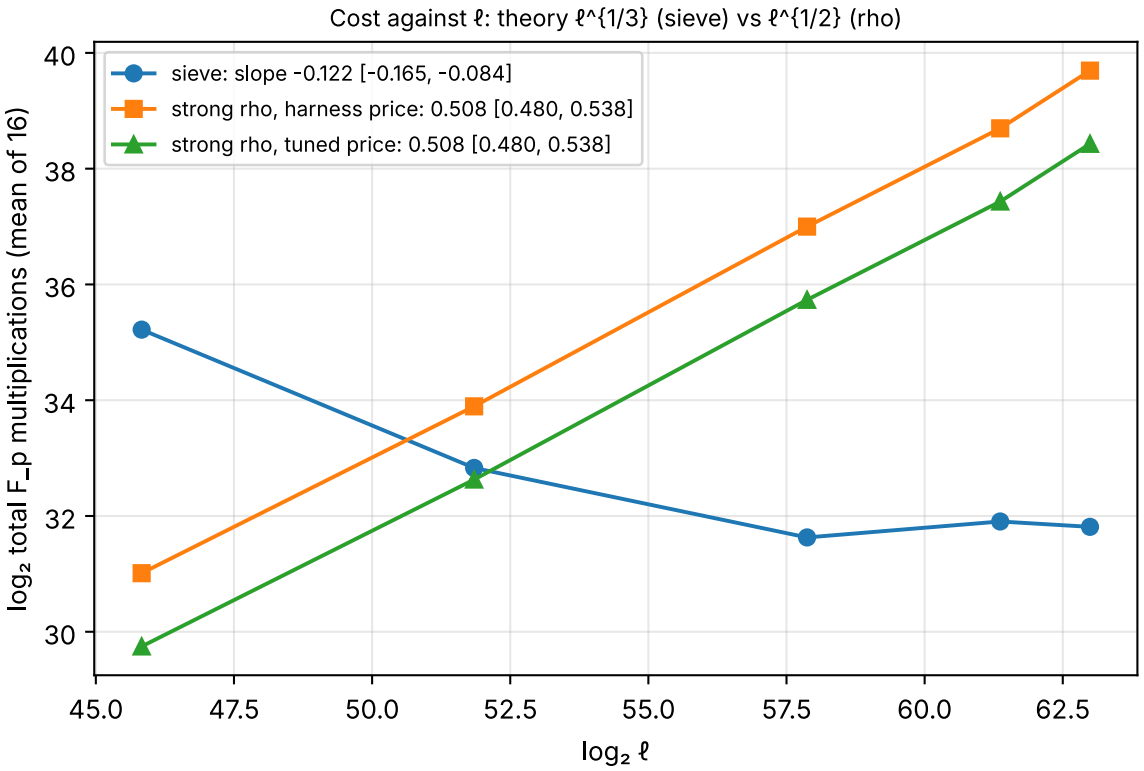


Figure 2. Mean total F_p multiplications against ℓ; fitted slopes in the legend (theory 1/3 sieve, 1/2 rho). Study jv_sieve_strong_rho_20261019, instances seeds 1–16 per size, 95% bootstrap intervals; source summary.json.

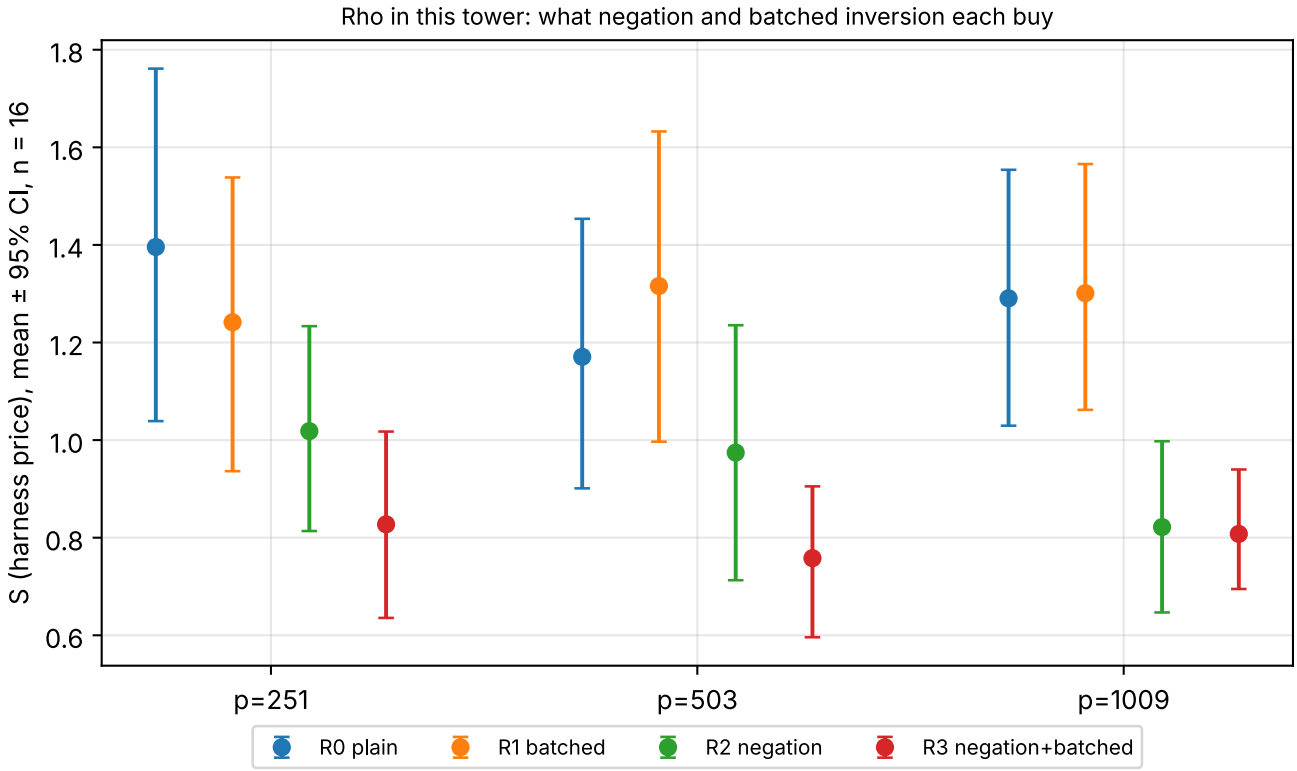


Figure 3. Rho factorial at p = 251, 503, 1009: negation map and batched inversion, harness price. Study jv_sieve_strong_rho_20261019, instances seeds 1–16 per size, 95% bootstrap intervals; source summary.json.

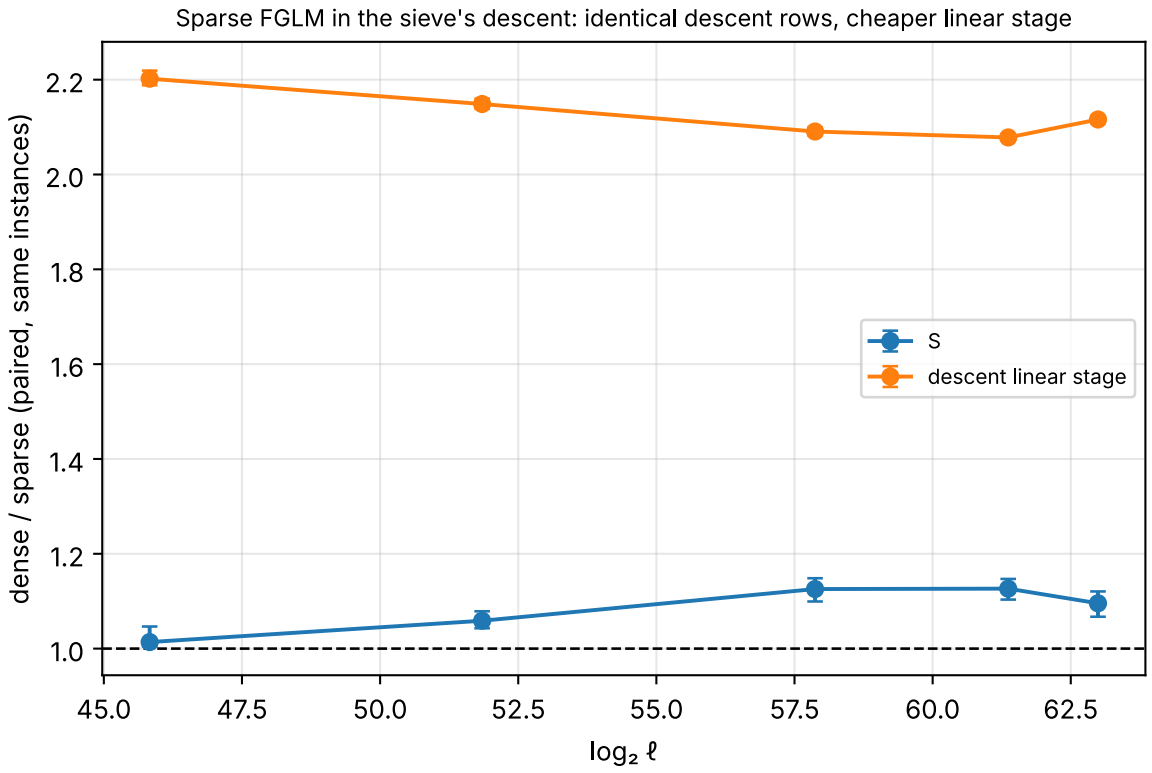


Figure 4. Dense over sparse FGLM in the sieve's descent, paired on the same instances. Study jv_sieve_strong_rho_20261019, instances seeds 1–16 per size, 95% bootstrap intervals; source summary.json.

T1. Sieve against the strong rho (R3), paired by instance

S, S⁺ in units of 331·√ℓ F_p multiplications. Ratios are sieve/rho: below 1 is the sieve cheaper. 95% bootstrap over instances.

p	$\log_2 \ell$	n	sieve S	sieve S⁺	rho S (harness / tuned)	rho S⁺ (harness / tuned)	S ratio, harness	S ratio, tuned	S⁺ ratio, harness	S⁺ ratio, tuned
251	45.8	16	15.28 [4.63, 34.6]	35.84 [11.9, 78.9]	0.827 / 0.344	0.975 / 0.492	18.47 [5.35, 46.6]	44.41 [12.9, 109]	36.76 [11.7, 88.6]	72.9 [22.9, 176]
503	51.8	16	0.3625 [0.264, 0.471]	1.195 [0.839, 1.59]	0.758 / 0.315	0.893 / 0.451	0.4783 [0.348, 0.64]	1.149 [0.842, 1.53]	1.338 [0.935, 1.82]	2.652 [1.85, 3.62]
1009	57.9	16	0.01954 [0.0161, 0.0231]	0.06346 [0.0554, 0.0711]	0.808 / 0.336	0.952 / 0.480	0.02419 [0.0185, 0.0311]	0.05814 [0.0444, 0.0741]	0.06666 [0.0532, 0.0837]	0.1321 [0.105, 0.165]
1511	61.4	16	0.007043 [0.00613, 0.00819]	0.0254 [0.0236, 0.0273]	0.780 / 0.325	0.920 / 0.464	0.009024 [0.00703, 0.012]	0.02169 [0.0169, 0.0289]	0.02762 [0.0221, 0.036]	0.05473 [0.0439, 0.0713]
1823	63.0	16	0.003761 [0.00323, 0.00434]	0.01605 [0.0148, 0.0174]	0.887 / 0.369	1.045 / 0.527	0.004241 [0.00324, 0.00575]	0.01019 [0.00779, 0.0138]	0.01535 [0.0125, 0.0196]	0.03043 [0.0249, 0.0384]

T2. §5a band: sieve/rho with native work priced at w GAE per

operation on every arm Layer A prices every non-multiplication counted operation including F_p additions; layer B excludes F_p additions. The sieve side adds a bound for F4/linear-algebra additions it does not emit (one per multiplication outside the sieve and enumeration). A ratio is a rank only where its whole interval is on one side of 1 at every weight and in all four columns.

p	w	A, harness	A, tuned	B, harness	B, tuned	ranked at every weight
251	0.0	18.5 [5.4, 46]	44.4 [13, 1.1e+02]	18.5 [5.3, 45]	44.4 [13, 1.1e+02]	no
	0.1	126 [44, 2.9e+02]	138 [48, 3.1e+02]	15.9 [5.9, 35]	21.1 [8, 47]	
	1.0	142 [50, 3.3e+02]	144 [50, 3.3e+02]	14.4 [6.3, 29]	15 [6.7, 31]	
503	0.0	0.478 [0.35, 0.64]	1.15 [0.84, 1.5]	0.478 [0.35, 0.63]	1.15 [0.84, 1.5]	no
	0.1	5.81 [4.1, 8]	6.35 [4.5, 8.6]	0.686 [0.49, 0.93]	0.907 [0.64, 1.2]	
	1.0	6.61 [4.7, 9.1]	6.68 [4.7, 9.1]	0.811 [0.56, 1.1]	0.844 [0.59, 1.2]	

1009	0.0	0.0242 [0.018, 0.031]	0.0581 [0.044, 0.075]	0.0242 [0.018, 0.031]	0.0581 [0.045, 0.074]	yes
	0.1	0.32 [0.25, 0.4]	0.35 [0.28, 0.44]	0.0307 [0.025, 0.038]	0.0405 [0.033, 0.05]	
	1.0	0.365 [0.29, 0.46]	0.368 [0.29, 0.46]	0.0345 [0.028, 0.043]	0.036 [0.029, 0.044]	
1511	0.0	0.00902 [0.007, 0.012]	0.0217 [0.017, 0.029]	0.00902 [0.007, 0.012]	0.0217 [0.017, 0.029]	yes
	0.1	0.136 [0.11, 0.18]	0.149 [0.12, 0.19]	0.0128 [0.01, 0.017]	0.0169 [0.014, 0.022]	
	1.0	0.156 [0.12, 0.2]	0.157 [0.13, 0.2]	0.015 [0.012, 0.019]	0.0156 [0.013, 0.02]	
1823	0.0	0.00424 [0.0032, 0.0058]	0.0102 [0.0078, 0.014]	0.00424 [0.0033, 0.0058]	0.0102 [0.0078, 0.014]	yes
	0.1	0.0761 [0.062, 0.097]	0.0832 [0.068, 0.11]	0.00708 [0.0058, 0.009]	0.00936 [0.0076, 0.012]	
	1.0	0.0869 [0.071, 0.11]	0.0878 [0.071, 0.11]	0.00879 [0.0073, 0.011]	0.00914 [0.0076, 0.011]	

T3. Rho factorial: what negation and batched inversion each buy (S, harness price; n per cell)

p	arm	n	steps/$\sqrt{\ell}$	S harness	S tuned	S^+ harness	wall s (1 core)
251	R0 plain	16	1.39 [1, 1.8]	1.4 [1, 1.8]	0.54 [0.41, 0.68]	1.59 [1.2, 2]	4
251	R1 batched	16	1.24 [0.94, 1.5]	1.24 [0.94, 1.5]	0.517 [0.39, 0.64]	1.41 [1.1, 1.8]	3
251	R2 negation	16	1.01 [0.81, 1.2]	1.02 [0.81, 1.2]	0.394 [0.31, 0.48]	1.2 [0.96, 1.5]	3
251	R3 negation + batched	16	0.822 [0.63, 1]	0.827 [0.64, 1]	0.344 [0.27, 0.42]	0.975 [0.74, 1.2]	2
503	R0 plain	16	1.17 [0.91, 1.5]	1.17 [0.9, 1.5]	0.453 [0.35, 0.56]	1.33 [1, 1.7]	26
503	R1 batched	16	1.32 [1, 1.6]	1.32 [1, 1.6]	0.548 [0.41, 0.68]	1.5 [1.1, 1.9]	23
503	R2 negation	16	0.971 [0.71, 1.2]	0.974 [0.71, 1.2]	0.377 [0.27, 0.48]	1.15 [0.84, 1.5]	24
503	R3 negation + batched	16	0.757 [0.59, 0.91]	0.758 [0.6, 0.91]	0.315 [0.25, 0.38]	0.893 [0.7, 1.1]	14
1009	R0 plain	16	1.29 [1, 1.6]	1.29 [1, 1.6]	0.499 [0.4, 0.6]	1.47 [1.2, 1.8]	241
1009	R1 batched	16	1.3 [1.1, 1.6]	1.3 [1.1, 1.6]	0.542 [0.44, 0.65]	1.48 [1.2, 1.8]	186
1009	R2 negation	16	0.82 [0.65, 0.99]	0.822 [0.65, 1]	0.318 [0.25, 0.39]	0.968 [0.77, 1.2]	160
1009	R3 negation + batched	16	0.808 [0.69, 0.94]	0.808 [0.69, 0.94]	0.336 [0.29, 0.39]	0.952 [0.82, 1.1]	120
1009	R0 A/A repeat	16	1.25 [0.96, 1.6]	1.25 [0.95, 1.6]	0.482 [0.37, 0.6]	1.42 [1.1, 1.8]	229
1009	R3 A/A repeat	16	0.824 [0.57, 1.1]	0.823 [0.58, 1.1]	0.343 [0.24, 0.46]	0.971 [0.68, 1.3]	121
1511	R0 plain	8	1.14 [0.92, 1.4]	1.14 [0.91, 1.4]	0.44 [0.35, 0.53]	1.29 [1, 1.6]	722
1511	R3 negation + batched	16	0.781 [0.6, 0.96]	0.78 [0.6, 0.96]	0.325 [0.25, 0.4]	0.92 [0.71, 1.1]	373
1823	R0 plain	4	1.45 [0.82, 2.1]	1.45 [0.82, 2.1]	0.56 [0.32, 0.81]	1.65 [0.93, 2.4]	1609
1823	R3 negation + batched	16	0.887 [0.71, 1.1]	0.887 [0.7, 1.1]	0.369 [0.29, 0.45]	1.05 [0.83, 1.3]	760

T4. Dense against sparse FGLM in the sieve's descent (paired, same instances)

p	pairs	identical descent digest	identical outcome	S dense/sparse	linear stage dense/sparse	descent muls dense/sparse	sparse⟶dense fallbacks per run
251	16	16	16	1.014 [1, 1.05]	2.202 [2.19, 2.22]	1.286 [1.28, 1.29]	7.50
503	16	16	16	1.059 [1.04, 1.08]	2.149 [2.14, 2.16]	1.286 [1.28, 1.29]	3.75
1009	16	16	16	1.126 [1.1, 1.15]	2.091 [2.08, 2.1]	1.283 [1.28, 1.29]	1.75
1511	16	16	16	1.126 [1.1, 1.15]	2.078 [2.07, 2.08]	1.285 [1.28, 1.29]	1.19
1823	16	16	16	1.096 [1.07, 1.12]	2.116 [2.11, 2.12]	1.286 [1.28, 1.29]	0.44

T5. Exponent of total F_p multiplications in ℓ

quantity	slope [95%]	sizes	status
sieve	−0.122 [−0.165, −0.084]	251–1823	pre-registered; theory 1/3
strong rho (harness price)	0.508 [0.480, 0.538]	251–1823	pre-registered; theory 1/2 (instrument check)
sieve	0.052 [−0.015, 0.118]	1009–1823	**post hoc** (m = 9 regime), not in the protocol

strong rho	0.498 [0.420, 0.571]	1009–1823	**post hoc**

Local sieve exponents between adjacent mean sizes: −0.397, −0.199, 0.079, −0.057.

IC A/A (sparse, p = 1009, run twice): 16/16 instances with identical counts and digests.

Pre-registered kill rules: K1 fired at p = none; K2 narrowed at p = none.